PHY202 Practice Exam on chapter(s): 23, 24

1. $\varepsilon = 7.50 \times 10^{3}\,\mathrm{V} \;\because\; \varepsilon = B \ell v$

2. $I_s = 1.30 \times 10^{-3}\,\mathrm{A} \;\because\; \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$

3. $\varepsilon = 291.\,\mathrm{V} \;\because\; \varepsilon = N A B \omega \sin(\omega t), \sin(\omega t) = 1$

4. $E_{ind} = 0.842\,\mathrm{J} \;\because\; E_{ind} = \frac{1}{2} L I^2$

5. $L = 1.83 \times 10^{-4}\,\mathrm{H} \;\because\; L = N \frac{\Delta \Phi}{\Delta t} = \frac{\mu_0 N^2 A}{\ell}$

6. $\mathrm{emf} = 64.6\,\mathrm{V} \;\because\; \mathrm{emf} = -N \frac{\Delta \Phi}{\Delta t}$

7. $I_p = 2.47\,\mathrm{A} \;\because\; \frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$

8. $E_0 = 1.68 \times 10^{3} \frac{\mathrm{V}}{\mathrm{m}} \;\because\; I_{ave} = \frac{c \varepsilon_0 E_0^2}{2} = \frac{c B_0^2}{2 \mu_0} = \frac{E_0 B_0}{2 \mu_0}$

9. $I = 2.56\,\mathrm{A} \;\because\; I(t) = I_0 e^{-t/\tau}, \tau = \frac{L}{R}$

10. $\varepsilon = 13.6\,\mathrm{V} \;\because\; \varepsilon = -N \frac{\Delta \Phi}{\Delta t}$

11. $\tau = 5.49 \times 10^{-4}\,\mathrm{s} \;\because\; \tau = \frac{L}{R}$

12. $B_{max} = 1.40 \times 10^{-6}\,\mathrm{T} \;\because\; c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E}{B} = f \lambda$